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Numerical Analysis

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تاریخ: _____

دن: _____

Q1 Newton Forward Formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0$$

$$+ \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 + \cdots$$

Expanding the polynomial

$$y = y_0 + p\Delta y_0 + \frac{(p^2-p)}{2!}\Delta^2 y_0 + \frac{(p^3-3p^2+2p)}{6}\Delta^3 y_0$$

$$+ \frac{p^4-6p^3+11p^2-6p}{24}\Delta^4 y_0 + \cdots$$

Now take integration

$$I = h\int_0^n \left[y_0 + p\Delta y_0 + \frac{(p^2-p)}{2}\Delta^2 y_0 \right.$$

$$\left. + \frac{(p^3-3p^2+2p)}{6}\Delta^3 y_0 + \Delta^4 y_0 \frac{(p^4-6p^3+11p^2-6p)}{24} \right] dp$$

Now Integration step by step

1st term:-

$$\int_0^n \Delta y_0 \, dp = \Delta y_0 \left| p \right|_0^n = y_0[n-0] = \Delta y_0(n)$$

2nd term:-

$$\int_0^n p \Delta y_0 \, dp = \Delta y_0 \int_0^n p \, dp = \Delta y_0 \left| \frac{p^2}{2} \right|_0^n$$

$$= \frac{\Delta y_0}{2} \left[n^2 - 0 \right] = \frac{n^2 \Delta y_0}{2}$$

3rd term:-

$$\int_0^n (\Delta^2 y_0) \left(\frac{p^2 - p}{2} \right) dp = \frac{\Delta^2 y_0}{2} \int_0^n (p^2 - p) \, dp$$

$$= \frac{\Delta^2 y_0}{2} \left| \left| \frac{p^3}{3} \right|_0^n - \left| \frac{p^2}{2} \right|_0^n \right| = \frac{\Delta^2 y_0}{2} \left[\frac{n^3}{3} - \frac{n^2}{2} \right]$$

$$= \Delta^2 y_0 \left[\frac{n^3}{6} - \frac{n^2}{4} \right]$$

4th term

$$\Delta^3 y_0 \int_0^n \frac{(p^3 - 3p^2 + 2p)}{6} dp = \frac{\Delta^3 y_0}{6} \int_0^n (p^3 - 3p^2 + 2p) \, dp$$

$$= \frac{\Delta^3 y_0}{6} \left(\left| \frac{p^4}{4} \right|_0^n - 3 \left| \frac{p^3}{3} \right|_0^n + 2 \left| \frac{p^2}{2} \right|_0^n \right)$$

$$= \frac{\Delta^3 y_0}{6}\left[\frac{n^4}{4} - n^3 + n^2\right] = \Delta^3 y_0\left[\frac{n^4}{24} - \frac{n^3}{6} + \frac{n^2}{6}\right]$$

5th term:-

$$\int_0^n \frac{(p^4 - 6p^3 + 11p^2 - 6p)}{24}\Delta^4 y_0\, dp$$

$$= \frac{\Delta^4 y_0}{24}\left[\left|\frac{p^5}{5}\right|_0^n - 6\left|\frac{n^4}{4}\right|_0^n + 11\left|\frac{p^3}{3}\right|_0^n - 6\left|\frac{p^2}{2}\right|_0^n\right]$$

$$= \frac{\Delta^4 y_0}{24}\left[\frac{n^5}{5} - \frac{6(n^4)}{4} + \frac{11(n^3)}{3} - 6\left(\frac{n^2}{2}\right)\right]$$

$$= \Delta^4 y_0\left[\frac{n^4}{120} - \frac{n^4}{16} + \frac{11n^3}{7} - \frac{n^2}{8}\right]$$

Adding all terms

$$I = h\left[y_0 n + \Delta y_0\left(\frac{n^2}{2}\right) + \Delta^2 y_0\left[\frac{n^3}{6} - \frac{n^2}{4}\right]\right.$$

$$+ \Delta^3 y_0\left[\frac{n^4}{24} - \frac{n^3}{6} + \frac{n^2}{6}\right]$$

$$\left. + \Delta^4 y_0\left[\frac{n^5}{120} - \frac{n^4}{16} + \frac{11n^3}{72} - \frac{n^2}{8}\right]\right] \quad (A)$$

Taking n = 3 in eqn (1)

$$\int_{x_0}^{x_0+3h} f(x)\,dx = h\left[3y_0 + \frac{9}{2}\Delta y_0 + \frac{1}{2}\left[\frac{27}{3} - \frac{9}{2}\right]\Delta^2 y_0 + \frac{1}{6}\left[\frac{81}{4} - 27 + 9\right]\Delta^3 y_0\right]$$

$$= h\left[3y_0 + \frac{9}{2}(y_1 - y_0) + \frac{9}{4}(y_2 - 2y_1 + y_0) + \frac{3}{8}(y_3 - 3y_2 + 3y_1 - y_0)\right]$$

$$= \frac{3h}{8}\left[y_0 + 3y_1 + 3y_2 + y_3\right] \rightarrow (1)$$

$$\int_{x_0+3h}^{x_0+6h} f(x)\,dx = \frac{3h}{8}\left[y_3 + 3y_4 + 3y_5 + y_6\right]$$

$$\int_{x_0+6h}^{x_0+9h} f(x)\,dx = \frac{3h}{8}\left[y_6 + 3y_7 + 3y_8 + y_9\right]$$

So, Final formula is

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{3h}{8}\left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3 + y_6 + y_9 + \dots y_{n-1})\right]$$

Q2 $\int_0^4 \sqrt{x^2+1}\,dx$, $n=4$

$h = \frac{\text{upper limit} - \text{lower limit}}{n} = \frac{4-0}{4} = 1$

$h = 1$

x	0	1	2	3	4
y	1	$\sqrt{2}$	$\sqrt{5}$	$\sqrt{10}$	$\sqrt{17}$
	y_0	y_1	y_2	y_3	y_4

Simpson formula

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{3}\left[(y_0+y_4) + 4(y_1+y_3) + 2(y_2)\right]$$

$$\int_{x_0}^{4} \sqrt{x^2+1}\,dx = \frac{1}{3}\left[(1+\sqrt{17}) + 4(\sqrt{2}+\sqrt{10}) + 2(\sqrt{5})\right]$$

$$= 9.3004$$

Q2 Simpson 3/8 Rule

$$I_1 = \int_0^3 \sqrt{x^2+1}\, dx \approx \frac{3h}{8}\left[y_0 + 3y_1 + 3y_2 + y_3\right]$$

$$I_1 = \frac{3(1)}{8}\left[1 + 3(1.4142) + 3(2.2361) + 3.1623\right]$$

$$I_1 = \frac{3}{8}\left[1 + 4.2426 + 6.7083 + 3.1623\right]$$

$$I_1 = 5.6674$$

And for I_2

$$I_2 = \int_3^4 \sqrt{x^2+1}\, dx \approx \frac{h}{2}\left[y_3 + y_4\right]$$

$$I_2 = \frac{1}{2}\left[3.1623 + 4.1231\right]$$

$$I_2 = 3.6427$$

So

$$I = I_1 + I_2 = 5.6674 + 3.6427$$

$$I = 9.3102$$

Q3 $y' = 3x^2 + y$, $0 \le x \le 1$

$h = 0.1$, $y(0) = 4$

Euler Method

$$\frac{dy}{dx} = 3x^2 + y$$

$$\frac{dy}{dx} = f(x, y)$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{n+1} = y_n + h(3x_n^2 + y_n)$$

Iteration 1:-

$x_1 = x_0 + nh = x_0 + h = 0 + 0.1 = 0.1$

$y_1 = y_0 + (0.1)(3x_0^2 + y_0) = 4 + (0.1)(3(0)^2 + 4)$

$\boxed{y_1 = 4.4}$

Iteration 2:-

$x_2 = x_0 + 2h = 0 + 2(0.1) = 0.2$

$y_2 = y_1 + (0.1)(3x_1^2 + y_1) = 4.4 + (0.1)[3(0.1)^2 + 4.4]$

$\boxed{y_2 = 4.843}$

Iteration 3:-

$x_3 = x_0 + 3h = (0) + 3(0.1) = 0.3$

$y_3 = y_2 + (0.1)(3x_2^2 + y_2) = (4.843) + (0.1)[3(0.2)^2 + 4.843]$

$\boxed{y_3 = 5.3393}$

Iteration 4:-

$x_4 = x_0 + 4(h) = 0 + 4(0.1) = 0.4$

$y_4 = y_3 + h[3x_3^2 + y_3] = 5.3303 + (0.1)[3(0.3)^2 + 5.3303]$

$\boxed{y_4 = 5.900023}$

Iteration 5:-

$x_5 = x_0 + 5h = 0 + 5(0.1) = 0.5$

$y_5 = y_4 + hf(3x_4 + y_4) = 5.900023 + 0.1(3(0.4)^2 + 5.900023)$

$\boxed{y_5 = 6.5382153}$

Iteration 6:-

$x_6 = x_0 + 6h = 0 + 6(0.1) = 0.6$

$y_6 = y_5 + (0.1)(3x_5^2 + y_5) = 6.538253 + (0.1)(3(0.5)^2 + 6.538253)$

$\boxed{y_6 = 7.2670783}$

Iteration 7:-

$x_7 = x_0 + 7h = 0 + 7(0.1) = 0.7$

$y_7 = y_6 + (0.1)(3x_6^2 + y_0)$

$= 7.2670783 + (0.1)[3(0.6)^2 + 7.2670783]$

$\boxed{y_7 = 8.1017861}$

Iteration 8:-

$x_8 = x_0 + 8h = 0 + 8(0.1) = 0.8$

$y_8 = y_7 + h(3x_7^2 + y_7)$

$y_8 = 8.101786 + (0.1)[3(0.7)^2 + 8.101786]$

$\boxed{y_8 = 9.05896471}$

Iteration 9:-

$x_9 = x_0 + 8h = 0 + 9(0.1) = 0.9$

$y_9 = y_8 + h(3x_8^2 + y_8)$

$y_9 = 9.05896471 + (0.1)[3(0.8)^2 + 9.05896471]$

$\boxed{y_9 = 10.15686118}$

Iteration 10:-

$x_{10} = x_0 + 9h = 0.9$

$y_{10} = y_9 + h(3x_9^2 + y_9)$

$= 10.15686118 + (0.1)[3(0.9)^2 + 10.15686118]$

$\boxed{y_{10} = 11.41554773}$

Q4 $\frac{dy^2}{dx^2} - x^2\frac{dy}{dx} - 2xy = 1$

$y(0)=1, \quad y'(0)=1$

$h=0.1, \quad y(0.3)=?$

Differential equation can be written as

$y''(x) - x^2 y'(x) - 2xy - 1 = 0$

$y'' - x^2y' - 2xy - 1 = 0 \quad -(1)$

let $y' = z$

$y'' = z'$, replace in (1)

$z' - x^2 z - 2xy - 1 = 0$

$z' = x^2 z + 2xy + 1$

$f(x,y,z) = \frac{dy}{dx} = z$

$g(x,y,z) = \frac{dz}{dx} = x^2 z + 2xy + 1$

1st Iteration

$y_1 = y_0 + hf(x_0, y_0, z_0) = 1 + (0.1) f(0.1, 1$

$= 1 + (0.1)(1) = 1.1$

$\boxed{y_1 = 1.1}$

$z_1 = z_0 + hf(x_0, y_0, z_0) = 1 + (0.1)(0+0+1)$

$\boxed{z_1 = 1.1}$

$x_1 = x_0 + h = x_0 + (1)(0.1) = 0 + 0.1 = 0.1$

$\boxed{x_1 = 0.1}$

Iteration 2:

$y_2 = y_1 + hf(x_1, y_1, z_1) = 1.1 + (0.1) f(0.1, 1.1, 1.1)$

$y_2 = 1.1 + (0.1)(1.1)$

$\boxed{y_2 = 1.21}$

$z_2 = z_1 + hf(x_1, y_1, z_1) = 1.1 + (0.1) f(0.1, 1.1, 1.1)$

$z_2 = 1.1 + (0.1)(1.2310)$

$\boxed{z_2 = 1.2231}$

$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$

2nd iteration

$y_3 = y_2 + hf(x_2, y_2, z_2) = 1.21 + (0.1) f(0.2, 1.21, 1.2231)$

$y_3 = (1.21) + (0.1)(1.2231)$

$\boxed{y_3 = 1.33231}$

$z_3 = z_2 + hg(x_3, y_3, z_3) = 1.2231 + (0.1) g(0.2, 1.21, 1.2231)$

$= 1.2231 + (0.1)(4.008)$

$\boxed{z_3 = 1.6239}$

$x_3 = x_2 + h = 0.2 + 0.1 = 0.3$

So

$\boxed{y(0.3) = 1.33231}$

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Q5

$$\frac{dy}{dx} = 3x + y^2, \quad y(1) = 1.2$$

$$y(1.1) = ?$$

According to "single step" Rule of thumb.
If target point is very close to initial values, then we assume $n = 1$

$$h = \frac{x_n - x_0}{h} = \frac{1.1 - 1}{1} = 0.1$$

$$\boxed{h = 0.1}$$

For K_1 :-

$$K_1 = h f(x_n, y_n) = (0.1) f(x_0, y_0)$$

$$= (0.1) f(1, 1.2) = (0.1)[3(1) + (1.2)^2]$$

$$K_1 = 0.444$$

For K_2 :-

$$K_2 = h f(x_n + h, y_n + K_1) = (0.1) f(x_0 + h, y_0 + K_1)$$

$$= (0.1) f(1 + 0.1, 1.2 + 0.444)$$

$$= (0.1) f(1.1, 1.644)$$

$$K_2 = 0.60027$$

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For K_3:-

$K_3 = h f(x_n + h, y_n + K_2)$

$= (0.1) f(x_0 + 0.1, y_0 + K_2)$

$= (0.1) f(1 + 0.1, 1.2 + 0.60027)$

$K_3 = (0.1) f(1.1, 1.80027)$

$K_3 = 0.6540$

For K_4:-

$K_4 = h f(x_n + \frac{h}{2}, y_n + \frac{K_1}{3})$

$= (0.1) f(x_0 + \frac{0.1}{2}, y_0 + \frac{K_1}{3})$

$= (0.1) f(1 + 0.05, 1.2 + \frac{0.444}{3})$

$= (0.1) f(1.05, 1.7920)$

$K_4 = 0.5229$

Final K

$K = \frac{1}{6}(K_1 + 4K_4 + K_3)$

$K = \frac{1}{6}(0.444 + 4(0.5229) + 0.6540)$

$K = 0.5316$

So

$x_1 = x_0 + h = 1 + 0.1 = 1.1$

$y_1 = y_0 + K = 1.2 + 0.5316$

$\boxed{y_1 = 1.7316}$

$\boxed{y(1.1) = 1.7316}$